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RESEARCH

Metformin drives HIF-1α–mediated dual metabolic reprogramming to enhance $\gamma\delta$ T cell therapy in triple-negative breast cancer

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Abstract

Triple-negative breast cancer (TNBC) lacks effective targeted treatments, rendering $\gamma\delta$ T cell immunotherapy a promising therapeutic strategy. However, the function of these immune cells is often limited by exhaustion and immunosuppression. This study investigated whether metformin can enhance $\gamma\delta$ T cell-mediated immunity against TNBC. Results demonstrated that metformin increased the cytotoxicity, proliferation, and cytokine production of $\gamma\delta$ T cells while reducing their exhaustion markers. It differentially modulated cellular metabolism by enhancing oxidative phosphorylation (OXPHOS) and glycolysis in $\gamma\delta$ T cells while suppressing these pathways in cancer cells through AMPK-HIF1-α signaling. Metformin also upregulated stress ligands on tumor cells, thereby improving immune recognition. In chemoresistant models, metformin restored $\gamma\delta$ T cell function. Clinical data further showed that high AMPK activity and increased $\gamma\delta$ T cell infiltration were associated with improved patient survival. These findings indicate that metformin remodels immunometabolism and enhances tumor immunogenicity, supporting its potential as a combinatory agent in $\gamma\delta$ T cell-based immunotherapy for TNBC.

Keywords Metformin · $\gamma\delta$ T cells · TNBC · Metabolic reprogramming

Introduction

Triple-negative breast cancer (TNBC) is defined as a unique breast cancer subtype characterized by the absence of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2) expression [1]. This aggressive malignancy comprises 12–20% of breast cancer cases and demonstrates worse clinical outcomes than other subtypes [2, 3]. TNBC typically exhibits high histological grade, rapid proliferation kinetics, and elevated risk of early recurrence, with most events occurring within 3–5 years after diagnosis [2, 4].The disease exhibits a predilection for visceral metastases, particularly to the lungs and brain, distinguishing it from other breast cancer variants. Without therapeutic targets for endocrine or HER2-directed agents, systemic chemotherapy remains the cornerstone of TNBC management [5].

In recent years, immunotherapy has transformed cancer treatment paradigms. Immune checkpoint inhibitors (ICIs), such as agents targeting PD-1/PD-L1, demonstrated tumor regression in select TNBC patients, as evidenced by clinical trials [6, 7]. The KEYNOTE-522 trial further established pembrolizumab combined with chemotherapy as an effective regimen for early-stage TNBC [8, 9]. Adoptive cell therapy (ACT) has similarly shown therapeutic potential, with tumor-infiltrating lymphocyte (TIL) transfer mediating objective responses in metastatic breast cancer patients during preliminary investigations [10, 11]. These immunotherapeutic approaches collectively represent significant progress in TNBC management, with continued research likely to yield further improvements.

$\gamma\delta$ T cells constitute a distinct lymphocyte population that contributes substantially to antitumor immunity.

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Unlike conventional αβ T cells, they express $\gamma\delta$ T cell receptors that recognize antigens independently of MHC molecules [12, 13], enabling direct detection of tumor-associated ligands such as MICA/B and phosphoantigens like isopentenyl pyrophosphate (IPP) [12]. These cells mediate tumor cell lysis through cytotoxic effector functions. The Vγ9Vδ2 subset, predominant in human peripheral blood, demonstrates potent antitumor and antimicrobial activity. Studies have shown that these cells can be effectively expanded in vitro, emerging as compelling candidates for cancer immunotherapy [14, 15]. Their effector mechanisms include perforin/granzyme-mediated cytotoxicity, death ligand expression, and secretion of IFN-γ and TNF-α to orchestrate immune responses [13].

By combining innate-like rapid responsiveness with antigen-presenting capacity, $\gamma\delta$ T cells bridge the gap between innate and adaptive immunity while maintaining tissue surveillance in epithelial barriers. Within tumor microenvironments (TME), they counteract immunosuppressive networks [16], supporting their therapeutic potential. Clinical trials employing adoptively transferred in vitro-expanded $\gamma\delta$ T cells have reported reduced tumor burden and prolonged survival in select malignancies [17–19]. While further investigation is required, these findings underscore $\gamma\delta$ T cells' emerging role in oncologic therapeutics.

Metformin, a first-line anti-diabetic drug, exhibits notable antitumor effects mediated by AMP-activated protein kinase (AMPK) activation and mTOR signaling inhibition in tumor cells [20–22]. Interestingly, this effect appears to be context dependent, as metformin has also been reported to enhance effector T cell function [23, 24] while attenuating regulatory T cell (Treg) immunosuppression [25]. Synergistic effects with ICIs arise from improved immune cell infiltration and functionality [26, 27]. These observations support metformin as a potential adjunct to cancer immunotherapy. Its ability to modulate the TME for enhanced $\gamma\delta$ T cell efficacy in triple-negative breast cancer, however, requires further investigation.

Here, it is proposed that dual modulatory effects are exerted by metformin through the enhancement of $\gamma\delta$ T cell metabolism and effector function, coupled with concurrent reprogramming of tumor cell metabolism to increase susceptibility to immune attack. A systematic examination was conducted to evaluate the influence of metformin on $\gamma\delta$ T cell proliferation, exhaustion, and cytotoxicity, as well as its effects on tumor cell metabolic activity and immunogenic ligand expression. These findings were validated in vivo using TNBC xenograft models, including chemoresistant variants, with clinical relevance assessed through transcriptomic profiling and public database analysis. This study investigates whether $\gamma\delta$ T cell-mediated immunotherapy in TNBC could be enhanced through metformin repurposing via immunometabolic modulation.

Methods

Cell culture

Ethical approval for this study was obtained from the Ethics Committee of the First Affiliated Hospital, Sun Yat-sen University. Isolation of peripheral blood mononuclear cells (PBMCs) from healthy donors was performed using Ficoll-Paque PLUS (17,144,002, Cytiva)-based density gradient centrifugation. PBMCs were seeded at a concentration of 3×10^6 cells/mL and cultured in complete RPMI-1640 medium (C11875500BT, Gibco) with 10% heat-inactivated fetal bovine serum (h-FBS; 164,210, Procell) and stimulated with 10 μM Zoledronic Acid (Zol; ab-M2329, Abmole) supplemented with 100 IU/mL human recombinant IL-2 (CSBSJ-IL-2, Prosperich). Cells were cultured with IL-2 for 9–12 days. The MDA-MB-231 (CL-0150) and MDA-MB-468 (CL-0290) cell lines were obtained from Procell, while the MDA-MB-231-Luc (YC-D005-Luc-P) and MDA-MB-468-Luc (YC-D009-Luc-P) cell lines were purchased from Ubigene. All cell lines were maintained in Dulbecco's Modified Eagle Medium (C11995500BT, Gibco) with 10% FBS and 1% penicillin/streptomycin (15,140,122, Gibco). Culture conditions consisted of 37 °C incubation in a humidified atmosphere with 5% CO_2 tension.

Mice

All in vivo experiments received ethical approval from Sun Yat-sen University's Animal Ethics Committee prior to execution at the university's Animal Center facilities. The study utilized female BALB/c-nu mice aged 4–6 weeks, which were sourced from the Animal Experimental Center at Sun Yat-sen University's East Campus. Twenty-four h before tumor transplantation, all mice underwent total body irradiation at a dose of 200 cGy. Subsequently, MDA-MB-231 cells (0.1 mL containing 1×10^6 cells per mouse) mixed with Matrigel (1:1 volume ratio; 354,234, Corning) were subcutaneously injected to establish the xenograft model. On day 15 post-implantation, mice were randomly allocated into experimental groups based on comparable average tumor burden before treatment initiation. For investigation of metformin's impact on TNBC progression, animals were randomly divided into two experimental groups: the metformin (Met) treatment group and the control (PBS) group, receiving intraperitoneal (i.p.)

injections of metformin or PBS, respectively. To further examine the synergistic antitumor effects of metformin and $\gamma\delta$ T cells, we established four experimental groups: the PBS group, which received PBS via both tail vein (i.v.) and intraperitoneal (i.p.) injection; the metformin group, treated with metformin (i.p.) and PBS (i.v.); the $\gamma\delta$ T cell group, administered $\gamma\delta$ T cells (i.v.) and PBS (i.p.); and the metformin plus $\gamma\delta$ T cell combination group, which received $\gamma\delta$ T cells (i.v.) and metformin (i.p.). The $\gamma\delta$ T cell for injection were obtained from four healthy donors, with each injection batch utilizing cells from a single donor. The cells were expanded using the aforementioned method. Throughout the experiments, mice were monitored daily, and tumor volume was measured every 3 days using calipers. Body weight measurements were recorded at 6-day intervals. Tumor volume was calculated using the standard formula: Volume (mm^3) = (length × width^2)/2. By the guidelines of the Animal Center of Sun Yat-sen University, mice were humanely euthanized 36 days after tumor transplantation. Tumor tissues were collected and preserved for subsequent immunohistochemical analysis following sacrifice.

Cytotoxic assay

Assessment of $\gamma\delta$ T cell-mediated cytotoxicity was performed using luciferase-based assays, with MDA-MB-231-Luc, MDA-MB-468-Luc, MDA-MB-231/PTX-Luc, and MDA-MB-468/PTX-Luc cells serving as targets. To evaluate the effect of metformin on the cytotoxicity of $\gamma\delta$ T cells versus TNBC cells, metformin was added directly to the coculture system. In this system, $\gamma\delta$ T cells were cocultured with MDA-MB-231-Luc cells at E:T ratios of 2:1, 5:1, and 10:1, and with MDA-MB-468-Luc cells at E:T ratios of 1:1, 2:1, and 5:1. The cocultures were performed in 150 µL of culture medium in the white opaque plates for 24 or 48 h. To assess the antitumor effects of $\gamma\delta$ T cells following metformin treatment, $\gamma\delta$ T cells were pre-cultured for 48 h in a medium with metformin. These cells were then cocultured with MDA-MB-231-Luc cells at E:T ratios of 5:1 and 10:1, cocultured with MDA-MB-468-Luc cells at E:T ratios of 2:1 and 5:1 for an additional 48 h. To measure tumor cell killing after metformin treatment, MDA-MB-231-Luc and MDA-MB-468-Luc cells were treated with metformin for 24 h and subsequently cocultured with $\gamma\delta$ T cells at E:T ratios of 10:1 and 5:1, respectively, for 24 h. The cytotoxicity (%) was calculated using the following formula:

$$\text{Cytotoxicity} = [1 - (RLU_{treat} - RLU_{\beta kr})/(RLU_{sp} - RLU_{\beta kr})] \times 100\%$$

where RLU_{treat}, RLU_{sp}, and $RLU_{\beta kr}$ represent luminescence values of treated, spontaneous release, and background controls, respectively.

Intracellular cytokine assays

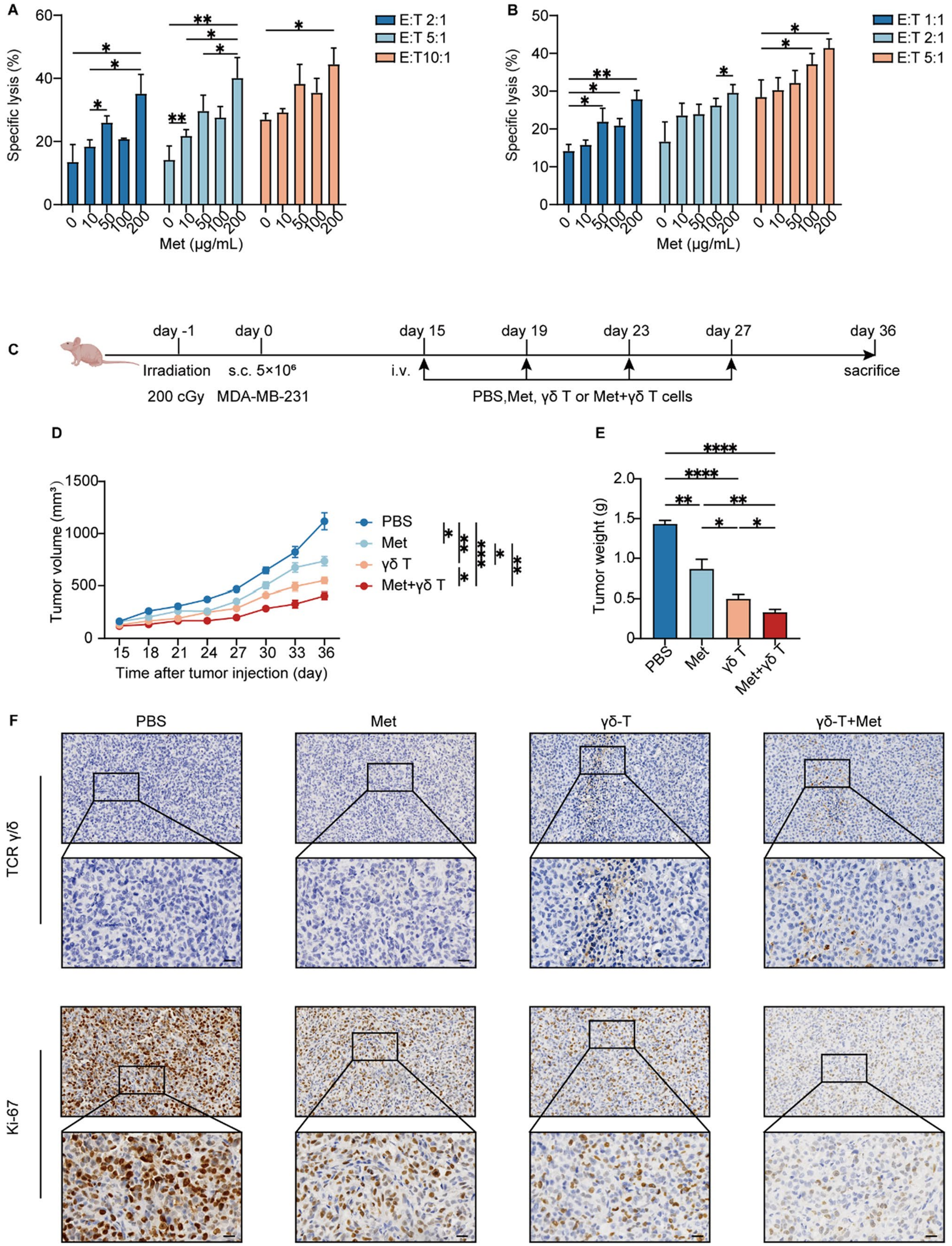
Investigation of metformin's effect on cytotoxic cytokine production was conducted by culturing $\gamma\delta$ T cells with the compound for 24 h. Subsequent flow cytometry was employed to assess the expression profiles of granzyme A, granzyme B, perforin, IFN-γ, and TNF-α. Cell stimulation was performed 4 h before analysis using PMA (100 ng/mL; P1585, Sigma-Aldrich) and ionomycin (1 µg/mL; ALX-450–006-M001, Enzo), with concurrent inhibition of cytokine secretion by brefeldin A (BFA; 10 µg/mL; B7651, Sigma-Aldrich). In the coculture assay, following a 48-h pretreatment with metformin (or vehicle control), $\gamma\delta$ T cells were cocultured with MDA-MB-231 target cells at a 10:1 E:T ratio for a 2-h incubation period. BFA was introduced to the culture system 4 h prior to analysis to facilitate intracellular protein accumulation. Subsequent flow cytometric analysis was performed to quantify expression levels of granzyme A, granzyme B, and perforin.

CCK-8 assay

The impact of metformin on cellular proliferation was assessed using CCK-8 assays in MDA-MB-231 and MDA-MB-468 breast cancer cell lines, as well as in $\gamma\delta$ T cells. Cell suspensions of MDA-MB-231 (5000 cells/well) and MDA-MB-468 (1×10^4 cells/well) were prepared and seeded into 96-well plates. After allowing for cell attachment, metformin treatment was initiated and maintained for a 48-h incubation period. $\gamma\delta$ T cells were expanded in culture for 12 days prior to experimentation. Subsequently, cells were seeded at a density of 1×10^6 cells per well in 96-well plates and subjected to metformin treatment for 72 h. After treatment, CCK-8 reagent (CK04, Dojindo) was added to the medium according to the manufacturer's instructions, and the plates were incubated for 3 h at 37 °C in a 5% CO_2 atmosphere. Optical density was quantified at 450-nm wavelength with a microplate reader to determine cellular responses.

Luciferase-based cell growth inhibition assay

Luciferase-based cytotoxicity assessment was conducted according to established protocols. In brief, $\gamma\delta$ T cells and target cells were cocultured in white opaque 96-well plates at specified E:T ratios, followed by incubation at 37 °C with 5% CO_2 for either 24 or 48 h. Luminescent signal detection was performed using the Steady-Glo® Luciferase Assay System (E2510, Promega) in strict accordance with the manufacturer's protocol.


A
Specific lysis (%)
Met (μg/mL)
E:T 2:1
E:T 5:1
E:T10:1
B
E:T 1:1
E:T 2:1
E:T 5:1
C
day -1
day 0
day 15
day 19
day 23
day 27
day 36
Irradiation
200 cGy
s.c. 5×10⁶
MDA-MB-231
i.v.
PBS,Met, γδ T or Met+γδ T cells
sacrifice
D
Tumor volume (mm³)
Time after tumor injection (day)
PBS
Met
γδ T
Met+γδ T
E
Tumor weight (g)
PBS
Met
γδ T
Met+γδ T
F
PBS
Met
γδ-T
γδ-T+Met
TCR γ/δ
Ki-67

◂**Fig. 1** Metformin synergizes with $\gamma\delta$ T cells to control TNBC. **A** Dose-dependent tumoricidal effects of $\gamma\delta$ T cells on MDA-MB-231-Luc cells at graded E:T ratios with metformin (Met) treatment, assessed by luciferase bioimaging. **B** Dose-dependent killing of MDA-MB-468-Luc cells by synergistic effect of $\gamma\delta$ T cells and metformin, measured via luciferase lysis assay. **C** Experimental timeline for xenograft studies: BALB/c-nu mice received injections of MDA-MB-231 cells (5×10^6 cells per mouse), followed by 15-day tumor establishment and randomization into treatment groups ($n = 6$). PBS or metformin was administered intraperitoneally every four days, 24 h before $\gamma\delta$ T cell administration through tail vein injection, covering four treatment cycles. **D** Longitudinal tumor volume measurements at indicated intervals. **E** Tumor weights were measured at the experimental endpoint (Day 36 post-inoculation) after humane euthanasia of the mice. **F** Immunohistochemical analysis of excised tumors showing $\gamma\delta$ T cell infiltration (TCR γ/δ) and proliferative index (Ki-67). Scale bar = 20 μm. All in vitro experiments were conducted with at least three biological replicates. Data presented as mean ± SEM. Statistical significance was determined by a two-tailed Student's t-test or two-way ANOVA. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$

Flow cytometry

For phenotypic characterization of $\gamma\delta$ T cells, the following mAbs targeting specific surface markers were employed: anti-CD3 (300,318, Biolegend), anti-CD4 (317,429, Biolegend), anti-CD8 (300,924, Biolegend), anti-TCR Vδ2 (331,418, Biolegend), anti-TCR Vδ2 (331,428, Biolegend), anti-PD-1 (329,906, Biolegend), anti-CTLA-4 (369,612, Biolegend), anti-TIM-3 (345,016, Biolegend), anti-LAG-3 (369,324, Biolegend), anti-Granzyme A (507,204, Biolegend), anti-Granzyme B (515,408, Biolegend), anti-perforin (353,316, Biolegend), anti-IFN-γ (506,507, Biolegend), and anti-TNF-α (502,926, Biolegend). To investigate the phenotypic characteristics of TNBC cells, the following mAbs were used: anti-BTN3A1 (130–124–880, Miltenyi Biotec), anti-MICA/B (742,322, BD Biosciences), anti-ULBP-1 (FAB1380A, R&D Systems), and anti-CD155 (337,614, Biolegend). Experimental protocols were executed following the manufacturer's specified guidelines. The apoptosis assay of TNBC cells was detected by Annexin V Apoptosis Detection Kit with PI (640,914, Biolegend). Viability was assessed using Fixable viability dye (65–0866-14, Thermo Fisher). Single fluorescence-stained compensation beads (424,602, Biolegend) were used for compensation adjustment. Multiparametric flow cytometry was conducted utilizing the following systems: FACS LSRFortessa™ (BD Biosciences), Attune® NxT (Thermo Fisher), and CytoFLEX (Beckman Coulter). Subsequent data analysis was performed using FlowJo v10.8.1 with standardized gating strategies.

Seahorse XF analysis

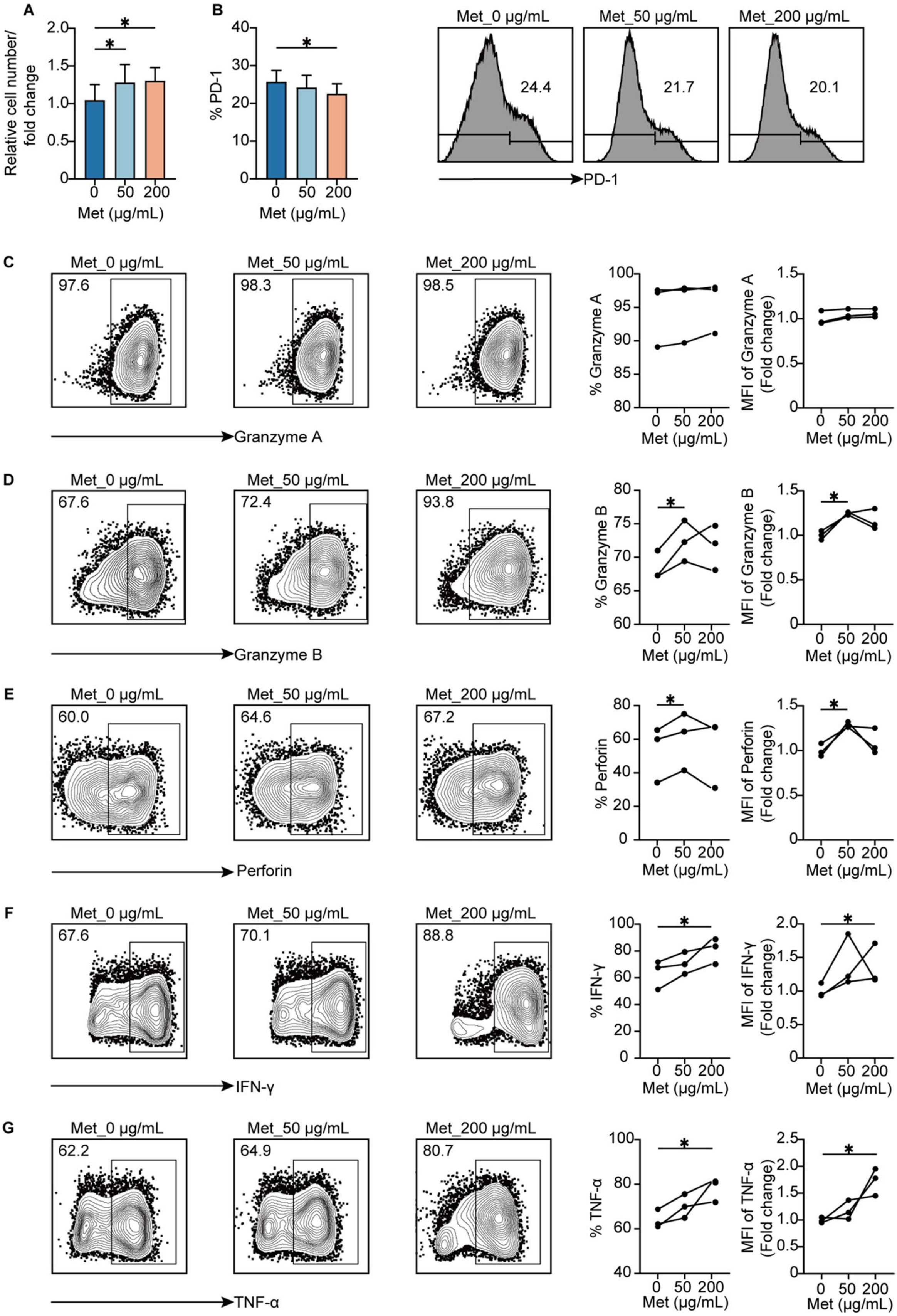
The real-time measurements of both oxygen consumption rate and extracellular acidification rate were carried out with a Seahorse XF96 Extracellular Flux Analyzer supplied by Agilent Technologies in accordance with the established protocol. Briefly, TNBC cells were plated in a Seahorse XF-96 cell culture microplate (Agilent) and permitted to adhere for 12 h under standard culture conditions prior to metabolic analysis, followed by the addition of culture medium containing metformin for 24 h. $\gamma\delta$ T cells were treated with metformin for 24 h prior to experimentation. Concurrently, cell culture microplates were coated with 0.2 mg/mL poly-L-lysine (ST508, Beyotime) for 30 min at room temperature. Following coating, cells were seeded onto the prepared plates and centrifuged at 300 × G to facilitate cell adhesion prior to subsequent experimental procedures. Subsequently, OCR and ECAR were measured using either XF DMEM (103,575–100, Agilent) or XF RPMI (103,576–100, Agilent) assay media, supplemented with 10 mM glucose, 2 mM L-glutamine, and 1 mM sodium pyruvate to maintain metabolic activity. OCR detection was added 1.0 μM oligomycin, 1.0 μM FCCP, and 0.5 μM rotenone plus antimycin A in order. ECAR detection was followed by 1.0 mM glucose, 1.0 μM oligomycin, and 50 mM 2-DG. Basal respiration, maximal respiration, and ATP production were calculated by OCR and glycolysis, glycolytic capacity and glycolytic reserve were calculated by ECAR.

SiRNA transfections

MDA-MB-231 and MDA-MB-468 cells underwent transfection utilizing Lipofectamine RNAiMAX (13,778,075, Thermo Fisher) under recommended conditions. BTN3A1 siRNA or control siRNA (HY-RS01673, MCE) was diluted in Opti-MEM I Reduced Serum Medium (31,985,062, Thermo Fisher) to a final concentration of 50 nM and complexed with Lipofectamine RNAiMAX reagent. siRNA and Lipofectamine RNAiMAX were combined in Opti-MEM I, followed by 15-min room temperature incubation for complex formation before 12-h cellular treatment at 37 °C. Following 12-h transfection, the medium was replaced with complete growth medium and cells were maintained for 48 h prior to analysis.

Lentivirus transfections

The lentivirus (Guangzhou Yuanjing Biotechnology) was packaged using a third-generation system for modifying


A
Relative cell number/ fold change
Met (μg/mL)
B
% PD-1
Met (μg/mL)
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
24.4
21.7
20.1
PD-1
C
97.6
98.3
98.5
Granzyme A
% Granzyme A
MFI of Granzyme A (Fold change)
D
67.6
72.4
93.8
Granzyme B
% Granzyme B
MFI of Granzyme B (Fold change)
E
60.0
64.6
67.2
Perforin
% Perforin
MFI of Perforin (Fold change)
F
67.6
70.1
88.8
IFN-γ
% IFN-γ
MFI of IFN-γ (Fold change)
G
62.2
64.9
80.7
TNF-α
% TNF-α
MFI of TNF-α (Fold change)

◀**Fig. 2** Metformin enhances the expansion and effector functions of *γδ* T cells **A** Healthy donor PBMCs were stimulated with Zol and maintained in IL-2-supplemented culture for 12 days, followed by 72-h metformin exposure of *γδ* T cells prior to CCK-8 proliferation assessment. **B** Flow cytometric evaluation of PD-1 surface expression on *γδ* T cells after 72-h metformin treatment. **C-G** Flow cytometry was employed to detect the release of granzyme A, granzyme B, perforin, IFN-γ, and TNF-α by *γδ* T cells following 24 h of metformin treatment. All experimental procedures included a minimum of three independent biological replicates. Data represent mean ± SEM; statistical significance determined by two-tailed Student's test. *$p < 0.05$

chemoresistant MDA-MB-231 and MDA-MB-468 cells. MDA-MB-231 and MDA-MB-468 cells were plated at densities of 1×10^6 and 2×10^6 cells per well, respectively, in 6-well plates. Transfection was performed at MOI 10 and 20, respectively, using a virus with polybrene (TR-1003-G, Merck Millipore) for 12 h. After PBS washing, cells were selected with puromycin (P8230, Solarbio) for 14 days, with luciferase assays confirming stable expression.

γδ T cell exhaustion model

γδ T cells were cultured for 9 days before experimentation. T Cell TransAct™ (130–111–160, Miltenyi Biotec) was used for restimulation at a titer of 1:500 following the manufacturer's established guidelines.

RNA-seq analysis

Following 12 days of in vitro expansion, *γδ* T cells were plated at 1×10^5 cells/mL in 48-well plates and exposed to metformin treatment for 48 h. Subsequent RNA isolation was performed using Trizol reagent (10,296,028, Invitrogen) according to the manufacturer's specifications. The RNA purification, cDNA synthesis, library preparation, and sequencing procedures were performed by Shanghai Majorbio Bio-pharm Biotechnology Co., Ltd. following established manufacturer protocols. Differential expression analysis was performed using DEGseq. Differentially expressed genes (DEGs) meet the dual thresholds of $|\log2FC| \geq 1$ and Benjamini–Hochberg corrected $p < 0.001$ were designated as significant differential expression events. Systematic functional annotation of DEGs was executed utilizing GO term enrichment and KEGG pathway analysis, complemented by hidden Markov model-based domain prediction via HMMER.

Immunohistochemistry

Formalin-fixed, paraffin-embedded (FFPE) tissue sections were processed for immunohistochemical staining via the HRP-polymer method. Following dewaxing, rehydration, and heat-mediated antigen retrieval in citrate buffer, endogenous peroxidase activity was quenched, and non-specific binding sites were blocked. Sections were then incubated overnight at 4 °C with primary antibodies against Ki-67 (ab15580, Abcam) and TCR γ/δ (TCR1061, Thermo Fisher), and then slices were rewarmed to room temperature. Next, the samples were incubated with anti-rabbit IgG HRP-linked antibody (7074S, CST) and goat anti-mouse IgG HRP (abs20039, Absin) at room temperature for 1 h. After hematoxylin nuclear staining, dehydration and clearing were performed, and in encapsulation, pictures were captured on a KF-PRO-020 (Kfbio).

Immunoblotting

Activation of AMPK and HIF-1α in TNBC and *γδ* T cells was evaluated via immunoblotting. Protein extraction was carried out after 4 h of metformin exposure using established methodologies. Briefly, cell lysates were prepared by incubating in ice-cold lysis buffer (PC101, EpiZyme) containing protease (C0001, TargetMol) and phosphatase inhibitors (C0004, TargetMol) for 20 min on ice. The lysate was centrifuged at 13,000 × g for 20 min at 4 °C. Samples were resolved on 10% SDS–polyacrylamide gels and electrotransferred to 0.45-μm nitrocellulose membranes (IPVH00010, Merck millipore). The membrane was blocked with 5% non-fat milk in TBST (JXF003, Jingxin Bio) for 1 h at room temperature and subsequently incubated overnight at 4 °C with primary antibodies targeting phospho-AMPKα (2535 T, CST), AMPKα (2793S, CST), HIF-1α (2178S, CST), and β-actin (4970S, CST). The membrane was then subjected to three consecutive washes with TBST for 5 min, incubated for 1 h at room temperature with Anti-rabbit IgG HRP-linked Antibody (7074S, CST) or goat anti-mouse IgG HRP (abs20039, absin), washed three times in TBST for 5 min, and detected using the Omni-ECL™ Femto Light Chemiluminescence Kit (SQ201, EpiZyme). The data were analyzed by ImageJ software (version 1.53).

Quantitative real-time qPCR

γδ T cells were treated with DMOG (89,464–63-1, MCE) for 24 h and RNA was extracted in Super FastPure Cell RNA Isolation Kit (RC102-01, Vazyme) and reverse-transcribed using HiScript IV RT SuperMix for RT−qPCR (R423-01, Vazyme). RT−qPCR was performed with SupRealQ Ultra Hunter SYBR qPCR Master Mix (Q713-02, Vazyme) on a QuantStudio 5 Real-Time PCR Systems(Thermo Fisher). Primer pairs for detection of human CXCR5 and SIX4 and internal ACTIN control are as follows:

CXCR5: forward, 5'-GAGAACCTGGAGGACCTG TTCTG-3';

reverse: 5'-TGGCAGGGCAGAGATGATTTT-3';

SIX4: forward, 5'-GTGGCAGCTTCACAAGGTAA-3';

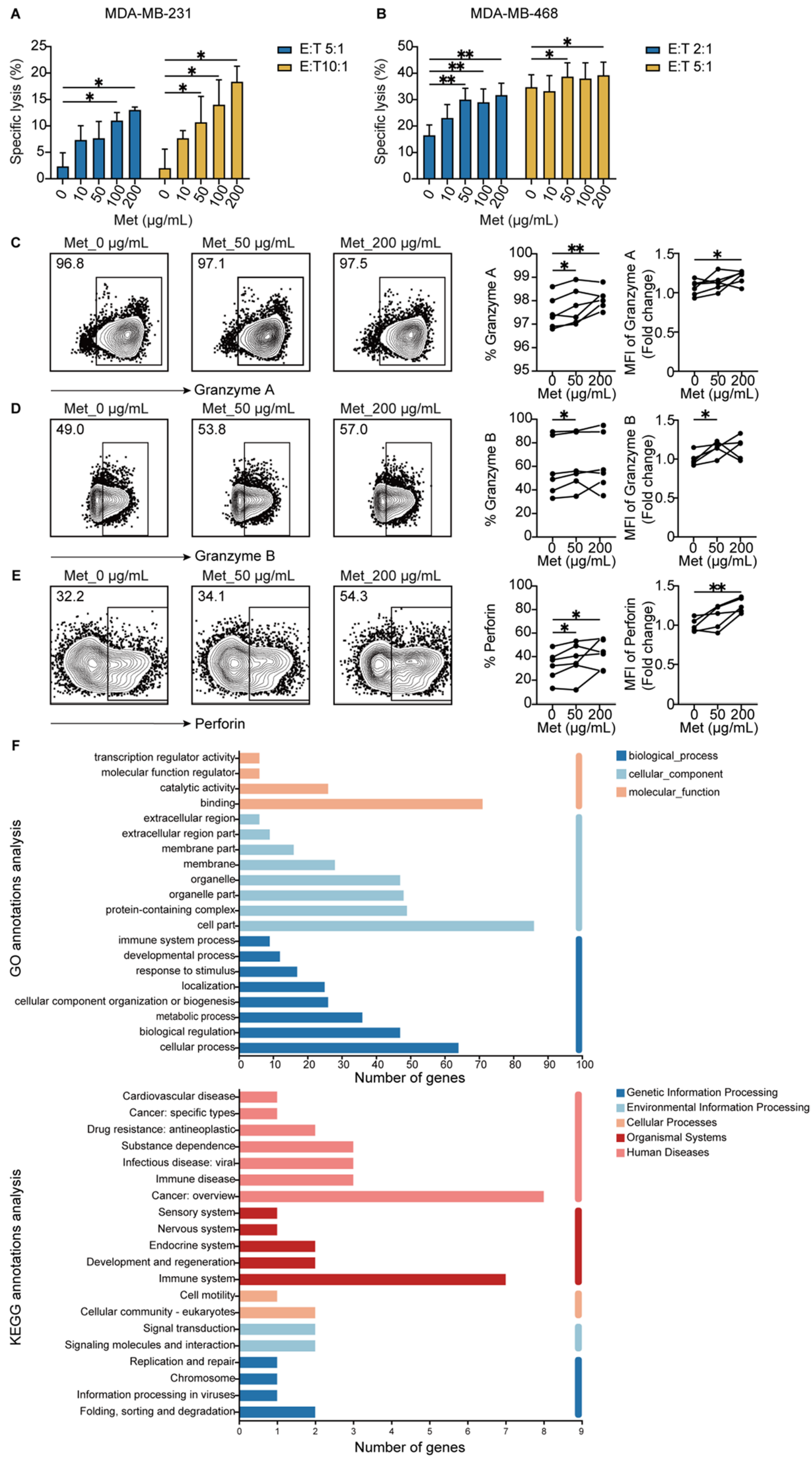


Fig. 3 Metformin potentiates $\gamma\delta$ T cell-mediated cytotoxicity against TNBC through enhanced effector functions and transcriptional reprogramming. **A, B** Tumoricidal capacity of $\gamma\delta$ T cells pretreated with metformin for 48 h against MDA-TNBC cells, assessed after 48 h of coculture using luciferase bioimaging. **C-E** Expression of effector cytokines in $\gamma\delta$ T cells pretreated with metformin for 48 h and subsequent to TNBC cell coculture, flow cytometry was employed for assessment. **F** Transcriptomic profiling of metformin-treated $\gamma\delta$ T cells: GO and KEGG annotations analysis of $\gamma\delta$ T cells differential genes. All experimental procedures included a minimum of three independent biological replicates. Quantitative data represent mean ± SEM; statistical significance determined by two-tailed Student's test. $^{*}p < 0.05$, $^{**}p < 0.01$

reverse, 5'-AGGCTCCTTTCCAAGCCTTC-3';

ACTIN: forward, 5'-AACTGGGACGACATGGAGAAAA-3';

reverse: 5'-GGATAGCACAGCCTGGATAGCA-3'.

Each reaction was performed in triplicates and normalized to β-actin gene expression. $2^{-\Delta\Delta CT}$ method was used to determine the relative expression.

Establishment of chemoresistant TNBC cell lines

Paclitaxel-resistant TNBC cell lines were generated by stepwise increasing paclitaxel concentrations. Parental MDA-MB-231 and MDA-MB-468 cells were first treated with 0.5 nM paclitaxel for 48 h, then recovered in drug-free medium. Each subculture cycle involved 48-h exposure to escalating paclitaxel doses followed by a drug-free recovery phase. Cells were cryopreserved at each passage to preserve intermediate resistance stages. If viability dropped severely at higher concentrations, the process was resumed using previously frozen cells from the last tolerated dose. For long-term maintenance, resistant cells were cultured with 5 nM paclitaxel. Paclitaxel IC50 was measured to confirm resistance. For validation, cells were seeded in 96-well plates at 5,000 or 10,000 cells per well, pre-cultured for 24 h, then treated with a paclitaxel concentration gradient for 96 h. Cell proliferation was assessed using CCK-8, and IC50 values were calculated via nonlinear regression.

GEIPA online database

The Gene Expression Profiling Interactive Analysis platform (GEPIA; http://gepia2.cancer-pku.cn/#index), a extensively employed bioinformatics resource for processing TCGA-derived tumor and matched normal transcriptomic datasets, was employed for prognostic evaluation. Survival analyses were conducted through this web-based resource to generate Kaplan–Meier curves.

GEO database

Transcriptomic data from the GSE76250 cohort were obtained from the Gene Expression Omnibus database (GEO), consisting of 165 triple-negative breast cancer samples and 33 normal tissues. Gene expression levels were quantified with Affymetrix HTA 2.0 arrays, with subsequent normalization through the Robust Multichip Analysis (RMA) pipeline. Target gene expression levels were obtained via GEO2R, followed by data cleaning and construction of an expression matrix. Differential expression between TNBC and normal tissues was analyzed using the Wilcoxon test ($p < 0.05$), with boxplots generated by ggplot2 in R.

Statistical analysis

Data are presented as mean ± SEM. Two-tailed Student's t-tests were used for comparisons between two groups, while one-way or two-way ANOVA with Tukey's post hoc test was applied for multiple comparisons. Mice were randomly allocated to experimental groups. All statistical analyses were performed using GraphPad Prism version 8.0, with statistical significance set at $p < 0.05$. The specific statistical tests and corresponding p-values for each experiment are provided in the figure legends.

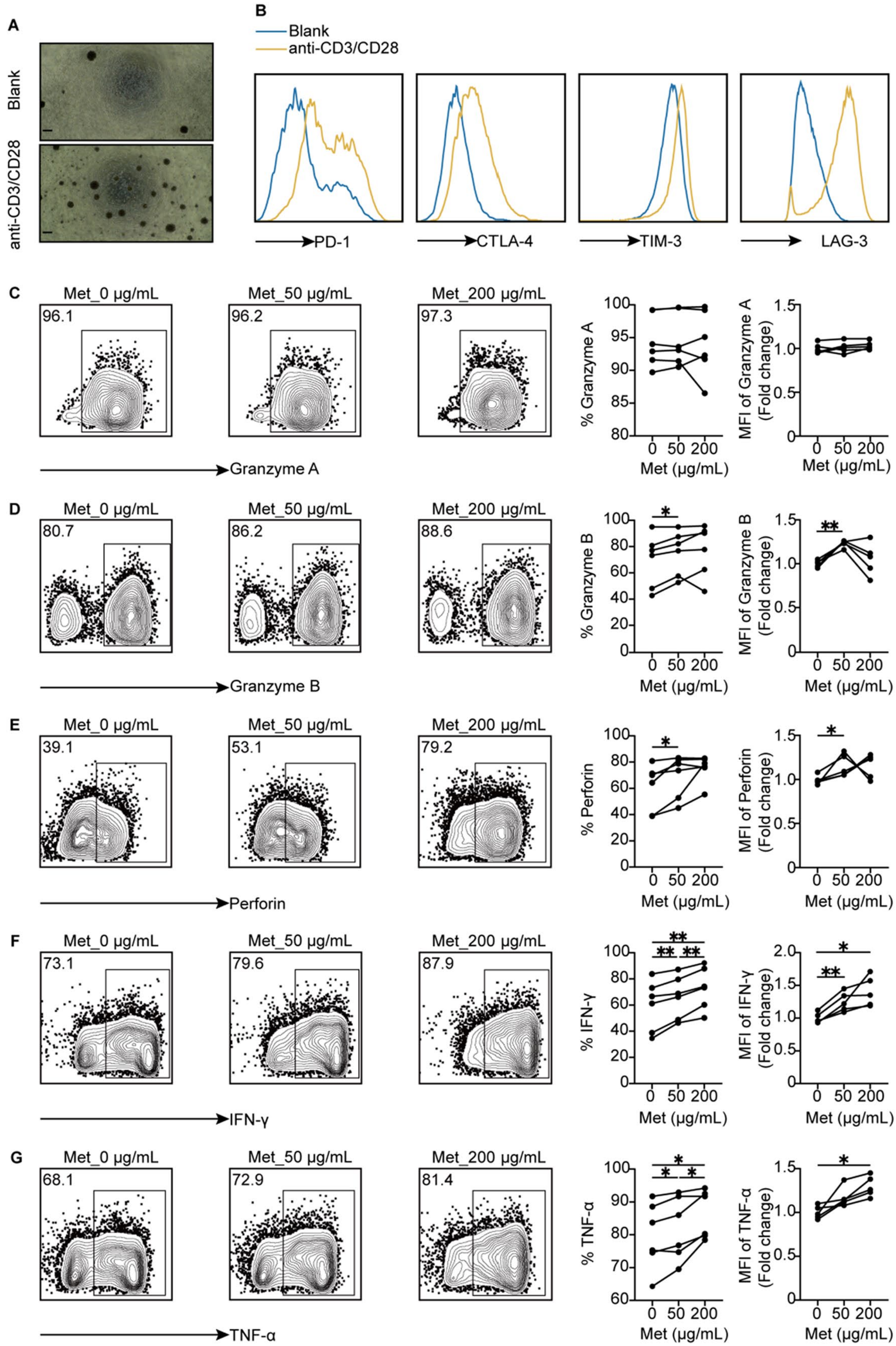
Results

Metformin synergize with γδ T cells to control TNBC

γδ T cells can be expanded to high purity with minimal $CD4^+$ and $CD8^+$ T cell contamination using zoledronate (Supplementary Fig. 1). To evaluate the antitumor activity of metformin in TNBC models, MDA-MB-231 and MDA-MB-468 cells were assessed for their responses to growth inhibition and apoptosis induction. Subsequent analyses monitored tumor growth suppression and survival extension in xenograft models (Supplementary Fig. 2A-G). The dual antitumor efficacy of metformin targeting TNBC was demonstrated both in vitro and in vivo, which aligns with prior reports of its direct tumor-suppressive properties [28–30].

To examine the combinatorial impact of γδ T cells and metformin on TNBC, MDA-MB-231 and MDA-MB-468 cells were cocultured with γδ T cells under graded metformin concentrations. Cytotoxic activity was quantified using a luciferase-based lysis assay (Fig. 1A, B). The cytotoxic efficacy of γδ T cells against malignant cells exhibited progressive enhancement with increasing dosage, indicating the capacity of metformin to potentiate γδ T cell cytotoxicity. These findings establish metformin as an enhancer of γδ T cell antitumor activity.

To evaluate the potential enhancement of γδ T cell-mediated antitumor responses by metformin in vivo, BALB/c-nu mice were subcutaneously inoculated with MDA-MB-231 cells. When tumors attained volumes between 100 and 150 mm^3, mice were randomly allocated to experimental cohorts administered with PBS (control) or γδ T cells. Metformin was administered intraperitoneally prior to the infusion of γδ T cells to evaluate its efficacy (Fig. 1C). Rapid tumor expansion was observed in the control group, while attenuated growth was demonstrated in all γδ T cell-treated groups. This tumor suppression was further potentiated


A
Blank
anti-CD3/CD28
B
Blank
anti-CD3/CD28
PD-1
CTLA-4
TIM-3
LAG-3
C
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
96.1
96.2
97.3
Granzyme A
% Granzyme A
MFI of Granzyme A (Fold change)
Met (μg/mL)
D
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
80.7
86.2
88.6
Granzyme B
% Granzyme B
MFI of Granzyme B (Fold change)
Met (μg/mL)
E
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
39.1
53.1
79.2
Perforin
% Perforin
MFI of Perforin (Fold change)
Met (μg/mL)
F
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
73.1
79.6
87.9
IFN-γ
% IFN-γ
MFI of IFN-γ (Fold change)
Met (μg/mL)
G
Met_0 μg/mL
Met_50 μg/mL
Met_200 μg/mL
68.1
72.9
81.4
TNF-α
% TNF-α
MFI of TNF-α (Fold change)
Met (μg/mL)

◂**Fig. 4** Metformin restores the antitumor capacity of $\gamma\delta$ T cells diminished by TCR activation-induced exhaustion. **A** Morphological characteristics of $\gamma\delta$ T cells following 72 h culture with or without anti-CD3/CD28 activation. Scale bar = 200 µm. **B** Surface levels of PD-1, CTLA-4, TIM-3, and LAG-3 on $\gamma\delta$ T cells were measured by flow cytometry after 72 h restimulation. **C-G** After 7 days of metformin exposure and restimulation, $\gamma\delta$ T cells were analyzed for production of granzyme A, granzyme B, perforin, IFN-γ, and TNF-α. All experimental procedures included a minimum of three independent biological replicates. Data represent mean ± SEM; statistical significance determined by two-tailed Student's test. $*p < 0.05$; $**p < 0.01$

by metformin pretreatment (Fig. 1D, E,Supplementary Fig. 3A). No significant differences in body weight were detected between $\gamma\delta$ T cell-treated and control mice (Supplementary Fig. 3B). Immunohistochemical analysis of resected tumors showed increased $\gamma\delta$ T cell infiltration in metformin-treated mice, together with reduced tumor cell proliferation, as indicated by Ki-67 expression (Fig. 1F). These findings indicate that metformin potentiates the in vivo antitumor activity of $\gamma\delta$ T cells.

Metformin promotes the proliferation and effector functions of *γδ* T cells

To assess $\gamma\delta$ T cell responsiveness, cells were exposed to escalating metformin concentrations, informed by prior evidence of metformin-enhanced $\gamma\delta$ T cell cytolytic cytokine secretion leading to improved immunotherapeutic efficacy [30]. Significant enhancement of $\gamma\delta$ T cell proliferative capacity was observed following metformin treatment (Fig. 2A), accompanied by reduced PD-1 expression (Fig. 2B). The effects of metformin on antitumor cytokine production were subsequently evaluated. Granzyme A expression in $\gamma\delta$ T cells remained unchanged, whereas low concentrations of metformin selectively increased granzyme B and perforin levels (Fig. 2C–E). Conversely, elevated metformin concentrations stimulated enhanced secretion of IFN-γ and TNF-α (Fig. 2F,G). The gating strategy is illustrated here with TNF-α as an example (Supplementary Fig. 4). These findings demonstrate the dose-dependent modulation of $\gamma\delta$ T cell effector functions by metformin.

Metformin potentiates *γδ* T cell-mediated cytotoxicity against TNBC through enhancing effector functions

To validate the influence of metformin pretreatment on $\gamma\delta$ T cell killing capacity, expanded $\gamma\delta$ T cells were treated with varying metformin concentrations prior to coculture with TNBC cells. A significant increase in $\gamma\delta$ T cell cytotoxicity was observed following metformin pretreatment (Fig. 3A, B). For assessment of cytotoxic cytokine production, TNBC cells were incubated with metformin-pretreated $\gamma\delta$ T cells preceding final assessment. Pre-incubation of $\gamma\delta$ T cells with metformin-enhanced granzyme A, granzyme B, and perforin expression in the coculture system (Fig. 3C-E). These findings demonstrate that the intrinsic antitumor function of $\gamma\delta$ T cells can be augmented by metformin.

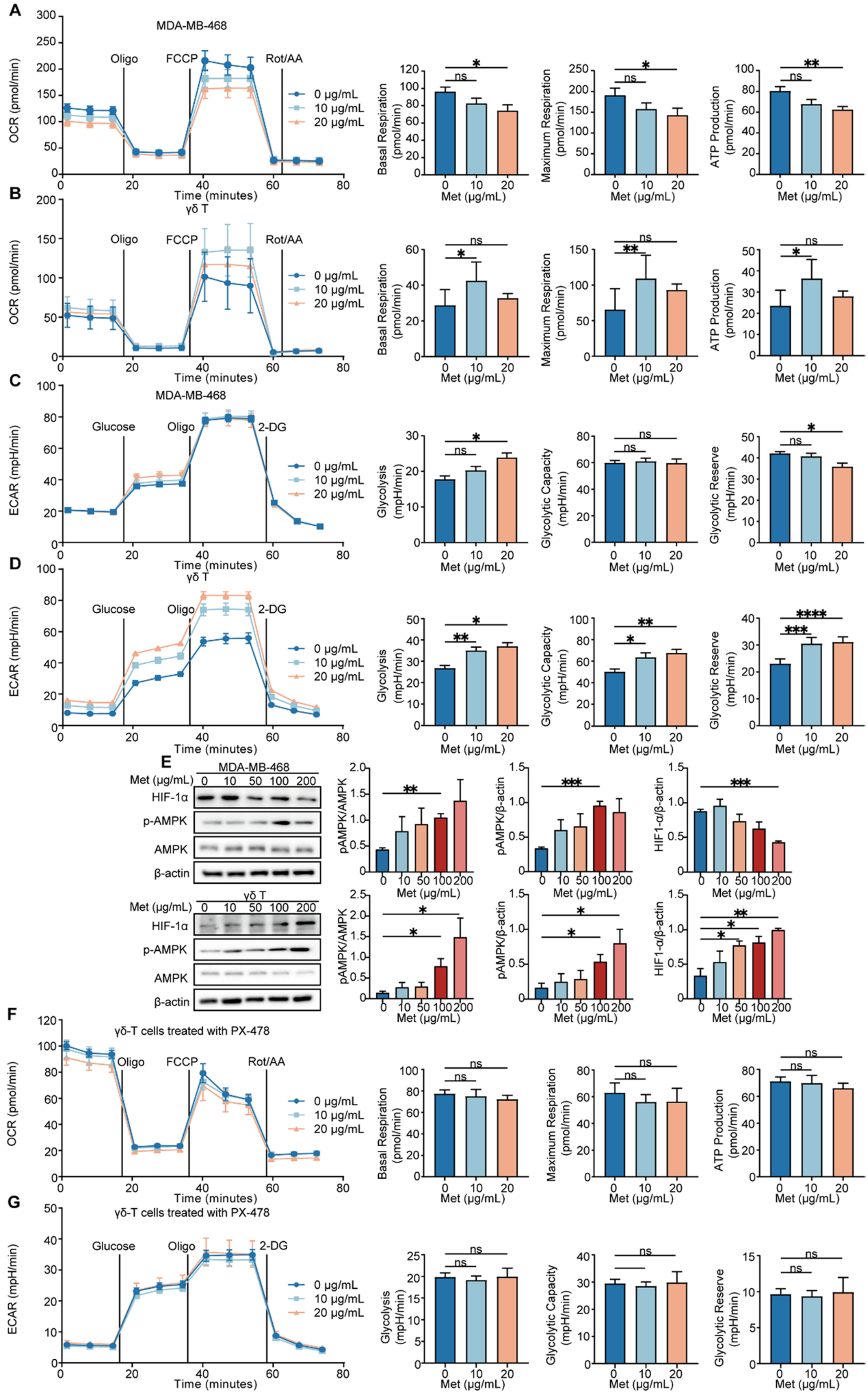
To elucidate the mechanistic basis for metformin-mediated potentiation of $\gamma\delta$ T cell antitumor efficacy, transcriptomic sequencing was conducted with subsequent differential gene expression profiling via gene ontology (GO) and kyoto encyclopedia of genes and genomes (KEGG) enrichment analyses. The optimal intervention timepoint was identified as day 9 based on expansion kinetics monitored prior to metformin treatment. GO analysis revealed that $\gamma\delta$ T cell function is predominantly altered by metformin through regulation of molecular binding, cellular components, and biological processes. KEGG pathway analysis further demonstrated pronounced effects of metformin on substance dependence, infectious diseases, immune disorders, cancer pathways, and immune system regulation (Fig. 3F).

Among the differentially expressed genes, CXCR5 upregulation enhances immune cell homing to lymphoid tissues. CXCR5 binding with its ligand CXCL13 triggers downstream signaling through PI3K, AKT, and ERK/MAPK pathways, which drive cellular proliferation and differentiation. Immune cells expressing high CXCR5 levels demonstrate increased cytotoxic activity, as evidenced by elevated granzyme B, TNF-α, and IFN-γ production [31, 32]. Elevated SIX4 expression enhances STING pathway activation. Studies demonstrate that SIX4 increases $CD8^{+}$T cell infiltration into tumors, thereby improving anti-PD-1 immunotherapy response [33]. TERC maintains telomere length and genomic stability while also modulating immune responses via NF-κB pathway activation and subsequent upregulation of proinflammatory cytokines, including TNF-α [34, 35].

These results demonstrate that metformin augments the tumoricidal capacity of $\gamma\delta$ T cells toward triple-negative breast cancer through the upregulation of cytotoxic mediators and the modulation of key immune pathways governed by CXCR5, SIX4, and TERC.

Metformin restores the antitumor capacity of *γδ* T cells diminished through TCR activation-induced exhaustion

Functional impairment and exhaustion of T cells in the TME arise from chronic TCR activatione, manifesting as attenuated effector responses, impaired proliferation, and elevated expression of inhibitory surface markers [36]. The exhausted state impairs antitumor immunity in cancer [37]. To determine whether metformin could restore the tumoricidal efficacy of exhausted $\gamma\delta$ T cells in the TME, an in vitro exhaustion model was established by inducing exhaustion with anti-CD3/CD28 activators (Fig. 4A). Three days after TCR


A
MDA-MB-468
Oligo
FCCP
Rot/AA
0 μg/mL
10 μg/mL
20 μg/mL
OCR (pmol/min)
Time (minutes)
Basal Respiration (pmol/min)
Maximum Respiration (pmol/min)
ATP Production (pmol/min)
Met (μg/mL)
ns
B
γδ T
C
Glucose
2-DG
ECAR (mpH/min)
Glycolysis (mpH/min)
Glycolytic Capacity (mpH/min)
Glycolytic Reserve (mpH/min)
D
E
Met (μg/mL) 0 10 50 100 200
HIF-1α
p-AMPK
AMPK
β-actin
pAMPK/AMPK
pAMPK/β-actin
HIF1-α/β-actin
F
γδ-T cells treated with PX-478
G

◂**Fig. 5** Metformin exerts distinct effects on OXPHOS and glycolysis in $\gamma\delta$ T cells versus TNBC cells through HIF-1α. **A**, **B** OCR profiles of TNBC cells and $\gamma\delta$ T cells following 24-h metformin treatment, measured by Seahorse XF metabolic flux analysis. OCR curves were assessed under basal conditions and after adding the indicated mitochondrial inhibitors: oligomycin, FCCP, and rotenone/antimycin. **C**, **D** ECAR of TNBC cells and $\gamma\delta$ T cells were quantified after 24 h of metformin exposure, with measurements taken during basal states and following sequential administration of glucose, oligomycin, and 2-DG. **E** Representative immunoblots showing AMPK signaling in TNBC and $\gamma\delta$ T cells after 4 h exposure to increasing metformin doses. β-actin was employed for normalization. **F**, **G** OCR and ECAR profiles of $\gamma\delta$ T cells following 24-h metformin and PX-478 treatment, measured by Seahorse XF metabolic flux analysis. All experimental procedures included a minimum of three independent biological replicates. Data represent mean ± SEM; statistical significance determined by two-tailed Student's test. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, ns, not significant. PX-478: HIF-1α Inhibitor

restimulation, a pronounced increase in immune checkpoint molecules PD-1, CTLA-4, LAG-3 and TIM-3 was observed in $\gamma\delta$ T cells (Fig. 4B). Metformin treatment increased the expression of granzyme B, perforin, IFN-γ, and TNF-α in these exhausted $\gamma\delta$ T cells, while granzyme A levels remained unchanged (Fig. 4C–G). The results suggest that metformin can recover the antitumor capacity of $\gamma\delta$ T cells diminished through TCR activation-induced exhaustion.

Metformin differentially modulates OXPHOS and glycolysis activity in γδ T cells and tumor cells through HIF-1α

Since metformin was found to inhibit TNBC cell growth while promoting $\gamma\delta$ T cell proliferation and effector functions, we postulated that metformin might exert distinct regulatory effects on glucose metabolism in $\gamma\delta$ T cells versus TNBC cells, possibly potentiating the tumor-killing capacity mediated by $\gamma\delta$ T cells. Using a Seahorse XF metabolic analyzer, comprehensive bioenergetic profiling of both TNBC cells and $\gamma\delta$ T cells was conducted. Metabolic flux analysis measured oxygen consumption rates (OCR) and extracellular acidification rates (ECAR) following metformin exposure. In TNBC cells, higher metformin concentrations progressively suppressed OXPHOS activity, reflected by reduced basal respiration, maximal respiration, and ATP generation (Fig. 5A and Supplementary Fig. 5A), whereas $\gamma\delta$ T cells exhibited substantial OXPHOS enhancement (Fig. 5B). TNBC cells exhibited dose-dependent increases in glycolysis following metformin treatment, consistent with the Warburg effect, while maintaining stable glycolytic capacity but showing a marked reduction in glycolytic reserve (Fig. 5C and Supplementary Fig. 5B). Conversely, $\gamma\delta$ T cells showed elevated glycolysis, glycolytic capacity, and glycolytic reserve (Fig. 5D). These findings suggest that metformin suppresses glucose metabolism in TNBC cells while boosting the metabolic activity in $\gamma\delta$ T cells.

To investigate underlying mechanism of metformin's distinct effects on glucose metabolism in TNBC cells and $\gamma\delta$ T cells, we assessed the expression of hypoxia-inducible factor-1α (HIF-1α), a key regulator of OXPHOS and glycolysis, following AMPK pathway activation [38]. This transcription factor modulates not only glycolysis but also OXPHOS [39]. AMPK can promote glycolysis and ATP production through the HIF-1α/GLUT1 axis, thereby sustaining estradiol synthesis [40]. In contrast, in other contexts, AMPK activation may suppress HIF-1α expression by enhancing its ubiquitination and degradation [41].

Metformin treatment significantly increased AMPK phosphorylation in both cell types, but HIF-1α expression decreased in TNBC cells while increased in $\gamma\delta$ T cells (Fig. 5E). PX-478 is a HIF-1α inhibitor capable of abrogating metformin's metabolic effects [42]. In contrast to exclusive metformin treatment demonstrating increased glycolysis in $\gamma\delta$ T populations (Fig. 5B, D), the addition of PX-478 completely abolished metformin's modulatory effects on both OXPHOS and glycolysis in $\gamma\delta$ T cells (Fig. 5F, G). Simultaneously, to further investigate the role of HIF-1α activation, the HIF-1α activator DMOG was employed in this study to examine the relationship between HIF-1α activation and the transcriptional levels of CXCR5 and SIX4 [43].It was shown that elevated HIF-1α activation enhanced the transcription of CXCR5 and SIX4, indicating that HIF-1α might mediate homing via direct CXCR5 activation and regulate STING signaling through direct SIX4 activation (Supplementary Fig. 6). These results demonstrate that metformin modulates AMPK-driven metabolism via divergent HIF-1α regulation in TNBC and $\gamma\delta$ T cells.

Metformin sensitizes TNBC cells to γδ T cell cytotoxicity by immunomodulation of surface immune ligands

Considering metformin's synergistic enhancement of $\gamma\delta$ T cell-mediated TNBC cytotoxicity, we investigated whether it could modulate the expression of surface immune ligands associated with γδ T cell recognition. TNBC cells were pretreated with metformin prior to coculture with $\gamma\delta$ T cells. A dose-dependent increase in $\gamma\delta$ T cell cytotoxic activity was observed (Fig. 6A). Elevated BTN3A1 surface expression was found in TNBC cells following metformin treatment (Fig. 6B). The functional significance of BTN3A1 induction in metformin-augmented $\gamma\delta$ T cell tumoricidal activity was verified through siRNA-mediated knockdown experiments. Following confirmation of efficient BTN3A1 silencing (Supplementary Fig. 7 and Fig. 6C), suppression of the metformin-induced enhancement in $\gamma\delta$ T cell cytotoxicity

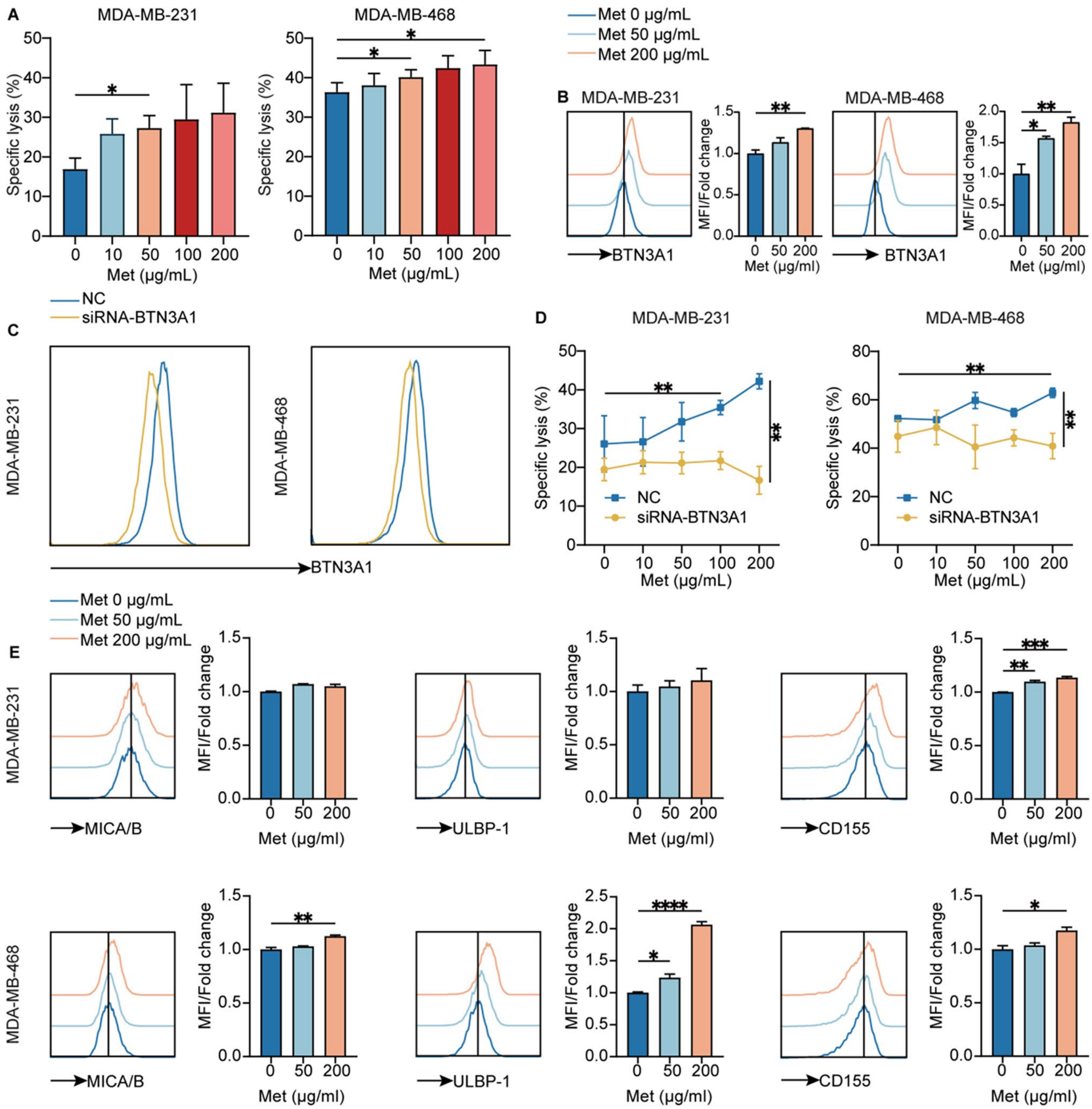


Fig.6 Metformin sensitizes TNBC cells to $\gamma\delta$ T cell cytotoxicity via immunomodulation of surface immune ligands. **A** Cytotoxic efficacy of $\gamma\delta$ T cells against TNBC cells following 24-h metformin pretreatment and subsequent 48-h coculture. **B** Expression of BTN3A1 on TNBC cells following 24-h treatment with metformin. **C** The validation of siRNA-mediated BTN3A1 knockdown. **D** Impact of BTN3A1 silencing on TNBC cell vulnerability to $\gamma\delta$ T cell cytotoxicity post 24-h metformin exposure. **E** Expression of MICA/B, ULBP-1, and CD155 on TNBC cells after 24 h of metformin treatment. All experimental procedures included a minimum of three independent biological replicates. Data represent mean ± SEM; statistical significance determined by two-tailed Student's test. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$

was documented (Fig. 6D). These results reveal the pivotal role of BTN3A1 in facilitating metformin-enhanced anticancer activity of $\gamma\delta$ T cells.

The cytotoxic potential of $\gamma\delta$ T cells is further governed by stimulatory receptors such as natural killer group 2 member D (NKG2D) and DNAX accessory molecule-1

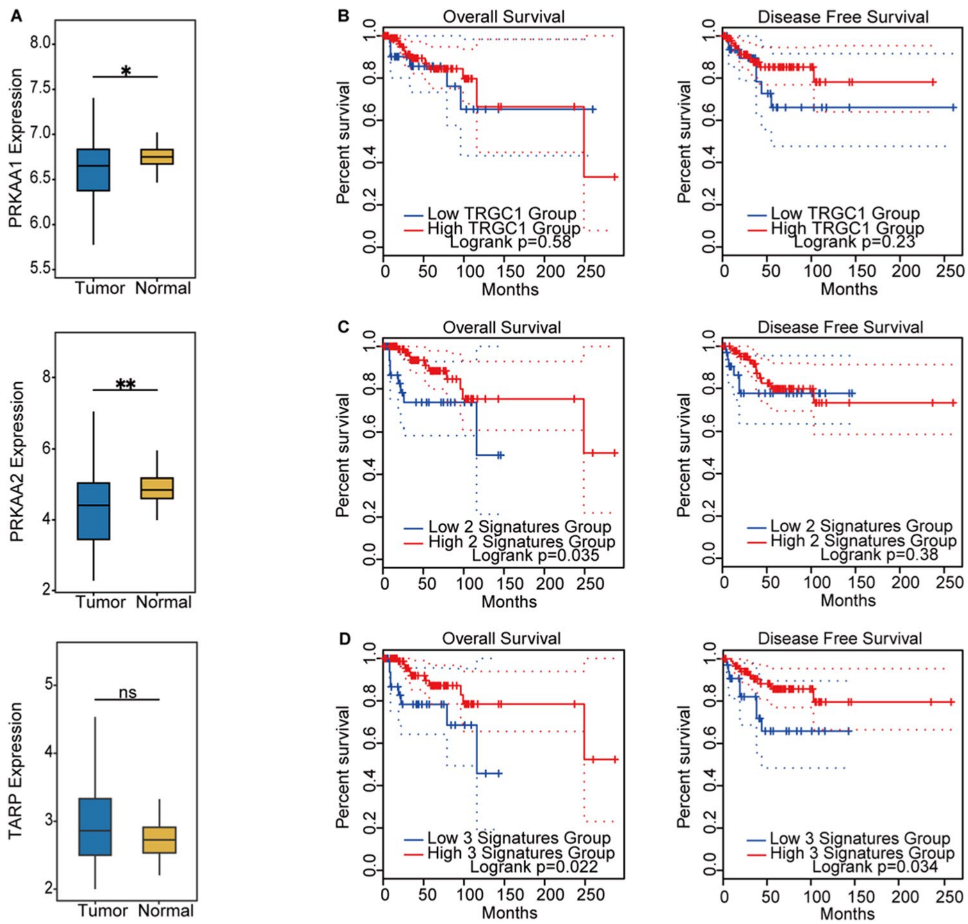


Fig. 7 AMPK pathway activation and $\gamma\delta$ T cell infiltration have positive prognostic implications for TNBC patients. **A** Analysis of the TCGA dataset (GSE76250) containing 165 TNBC specimens paired with 33 normal breast tissue controls demonstrated distinct mRNA expression profiles of PRKAA1, PRKAA2, and TRAP between cancerous and normal tissues, as visualized in the comparative box plot. **B-D** TCGA database analysis via GEPIA revealed that $\gamma\delta$ T cell infiltration combined with AMPK pathway activation correlates positively with OS and DFS in TNBC patients. **B** The correlation of TRGC1 mRNA expression with OS and DFS in TNBC patients. Low TRGC1 group: $n = 99$; high-signatures group: $n = 33$. **C** PRKAA1 and PRKAA2 were combined to analyze AMPK activation. The correlation of AMPK activation with OS and DFS of TNBC patients. Low-signatures group: $n = 100$; high-signatures group: $n = 34$. **D** The correlation of AMPK activation and $\gamma\delta$ T cell infiltration with OS and DFS in TNBC patients. Low-signatures group: $n = 100$; high-signatures group: $n = 34$. The Wilcoxon test determines the significance of mRNA expression profiles. The significance of survival from TCGA datasets was determined by the Log Rank test * $p < 0.05$, ** $p < 0.01$. ns, not significant

(DNAM-1), which recognize stress-induced ligands such as MICA/B, UL16-binding protein (ULBP)1–6, and CD155 on tumor cells [44, 45]. Importantly, the upregulation of these ligands enhances the cytotoxic potential of $\gamma\delta$ T cells [45, 46]. Flow cytometric analysis revealed that metformin treatment significantly upregulated CD155 expression in MDA-MB-231 cells, while MICA/B, ULBP-1, and CD155 were all elevated in MDA-MB-468 cells (Fig. 6E). These findings demonstrate that metformin enhances $\gamma\delta$ T cell antitumor activity through the upregulation of BTN3A1 expression to stimulate $\gamma\delta$ T cell activation, while concurrently enhancing NKG2D and DNAM-1 ligand presentation to fortify cytotoxic immunological synapse assembly.

Metformin enhances $\gamma\delta$ T cell-mediated cytotoxicity versus chemoresistant TNBC cells

TNBC represents a heterogeneous subtype that frequently exhibits resistance to standard chemotherapies including paclitaxel. Previous work demonstrated metformin's capacity to potentiate cisplatin's antitumor activity through radiation sensitive 51 (RAD51) downregulation, implicating DNA repair modulation as a potential mechanism for chemosensitization [47]. Both in vitro and in vivo studies confirmed complementary therapeutic effects when combining metformin with paclitaxel [48]. Further mechanistic investigation identified mTOR pathway inhibition as mediating metformin's enhancement of paclitaxel's anticancer efficacy [49].

To assess metformin's potential to amplify $\gamma\delta$ T cell-mediated killing of chemoresistant TNBC, paclitaxel-resistant MDA-MB-231/PTX and MDA-MB-468/PTX sublines were generated via progressive paclitaxel concentration escalation. A significantly higher paclitaxel IC50 was observed in resistant cell lines compared to parental cells (Supplementary Fig. 8A). For a comprehensive evaluation of synergistic effects, luciferase-expressing reporter cell lines were generated (Supplementary Fig. 8B) and cocultured with $\gamma\delta$ T cells across varying metformin concentrations, with cytotoxicity quantified through luminescence measurement. Dose-dependent increases in tumor cell death were observed with metformin treatment (Supplementary Fig. 9A), demonstrating enhanced $\gamma\delta$ T cell-mediated cytotoxicity contingent upon metformin administration. Dose-dependent inhibition in proliferation by metformin alone was demonstrated in both resistant cell lines (Supplementary Fig. 9B), along with apoptosis induction in chemoresistant TNBC models (Supplementary Fig. 9C). Subsequent investigation revealed significant upregulation of BTN3A1 expression in both resistant cell lines following metformin treatment(Supplementary Fig. 9D), paralleling the response in their non-resistant counterparts. Further examination of NKG2D and DNAM-1 ligand expression revealed cell line-specific modulation. Metformin treatment induced selective CD155 enhancement in MDA-MB-231/PTX cells, whereas it promoted a graded elevation of MICA/B, ULBP-1, and CD155 in MDA-MB-468/PTX cells (Supplementary Fig. 9E). These results indicate that metformin can counteract TNBC chemoresistance by reinstating $\gamma\delta$ T cell recognition through consistent modulation of immunogenic surface molecules.

Metformin application and $\gamma\delta$ T cell infiltration favor prognosis on TNBC

Metformin modulates cellular energy homeostasis through AMPK signaling [50]. The AMPK heterotrimer is composed of three distinct subunits, including the dominant catalytic α-subunit originating from PRKAA1 and PRKAA2 genes, alongside the regulatory β-subunit from PRKAB1 and PRKAB2 genes, plus the nucleotide-binding γ-subunit generated from PRKAG1 to PRKAG3 genes [51, 52], with the α-subunit being functionally predominant. Cellular AMPK activity exhibits positive correlation with the expression levels of these subunit genes [53].

To assess the influence of metformin-activated AMPK combined with $\gamma\delta$ T cell infiltration levels in TNBC patients, PRKAA1, PRKAA2, and TCR gamma alternate reading frame protein (TRAP) expression levels in TNBC versus normal tissues were examined using GEO datasets. PRKAA1 and PRKAA2 expression was significantly lower in TNBC tissues, whereas $\gamma\delta$ T cell infiltration showed no significant variation (Fig. 7A).

The TCGA database was further utilized to analyze the impact of AMPK pathway activation and $\gamma\delta$ T cell infiltration on the survival of TNBC patients. Elevated $\gamma\delta$ T cell infiltration was defined by TRGC1 expression levels within the top quartile, while the top quartile of combined PRKAA1 and PRKAA2 expression, referred to as the two signatures, served as a proxy for AMPK pathway activation. Additionally, the top quartile of combined TRGC1, PRKAA1, and PRKAA2 expression, termed the three signatures represented the convergence of AMPK pathway activation and high $\gamma\delta$ T cell infiltration. It was revealed that high $\gamma\delta$ T cell infiltration did not significantly improve OS or DFS in TNBC patients. However, AMPK pathway activation was associated with better OS, though no correlation was observed with DFS. When both high $\gamma\delta$ T cell infiltration and AMPK pathway activation were present, TNBC patients exhibited improved OS and DFS (Fig. 7B-D). These findings reveal that metformin-boosted AMPK signaling in tumor-infiltrating $\gamma\delta$ T cells contributes to enhanced TNBC clinical outcomes.

Discussion

Metformin enhances $\gamma\delta$ T cell-mediated antitumor immunity in TNBC by coordinating metabolic and immunomodulatory mechanisms. The drug directly increases $\gamma\delta$ T cell cytotoxicity by stimulating proliferation, suppressing exhaustion markers, and selectively altering effector molecule expression. Transcriptomic profiling identified upregulated immunomodulatory genes such as CXCR5 and SIX4, which facilitate lymphoid trafficking and STING pathway stimulation. Metformin induces metabolic compartmentalization by suppressing OXPHOS and glycolysis in tumor cells while enhancing mitochondrial respiration in γδ T cells, generating an energy gradient that promotes immune effector activity. Furthermore, metformin increases tumor cell expression of $\gamma\delta$ T cell-activating ligands, particularly BTN3A1 and stress-induced surface molecules. Clinical data correlate AMPK pathway activation with prolonged patient survival. These results position metformin as a clinically viable immunometabolic adjuvant for $\gamma\delta$ T cell therapies in attacking TNBC.

The combination of metformin and $\gamma\delta$ T cells demonstrated strong synergy in overcoming chemotherapy resistance in TNBC, showing potent cytotoxicity to drug-resistant TNBC cells in vitro. This synergistic effect depended on metformin concentration. Clinically, $\gamma\delta$ T cells offer advantages for ACT because they recognize tumors without MHC restriction, expand readily, and can be developed into off-the-shelf products targeting allogeneic tumors while minimizing GVHD risk. These properties make $\gamma\delta$ T cells a robust therapeutic platform. Our study demonstrates the clinical feasibility of combining metformin with γδ T cell immunotherapy for advanced TNBC management.

While our study demonstrates the synergistic antitumor effects of metformin and human $\gamma\delta$ T cells in TNBC xenograft models, several limitations must be acknowledged. First, our in vivo experiments were conducted in immunodeficient BALB/c-nu mice. This model allows efficient engraftment of human TNBC cells and adoptive transfer of human $\gamma\delta$ T cells without immune rejection, enabling us to isolate and study the direct effects of metformin on $\gamma\delta$ T cell function. However, this model does not recapitulate a complete immune system [54]. Actually, significant species-specific heterogeneity exists between human and murine $\gamma\delta$ T cells in terms of subset distribution, antigen recognition, and functional responses. Therefore, we did not assess the effects of metformin on human $\gamma\delta$ T cells in immunocompetent murine models. Future studies using humanized mouse models will be essential to evaluate the impact of metformin on human $\gamma\delta$ T cells within a more intact immune context. Second, although the expanded Vγ9Vδ2 T cell population achieved high purity (> 95%), a small fraction of residual non-$\gamma\delta$ T cells may remain. However, the use of MHC-mismatched cytotoxicity assays and strict gating on $CD3^{+}TCR\ V\delta 2^{+}$ cells in phenotypic analyses minimizes the potential contribution of contaminating αβ T cells. Together with the consistent enhancement of $\gamma\delta$ T cell effector functions observed across multiple assays, these findings support a $\gamma\delta$ T cell-intrinsic effect of metformin. Third, although we established this mechanistic framework in TNBC models, extending these findings to other solid tumors or metastatic settings will require validation across diverse tumor types and clinically representative platforms, particularly patient-derived xenografts and immunocompetent models. Addressing these gaps through multidisciplinary research could significantly progress metformin's clinical development as a metabolic adjuvant for $\gamma\delta$ T cell-based immunotherapy.

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Author contributions X. Q., H. Z., and M. L. contributed equally. X. Q. conceived the project, drafted the manuscript, and performed the experiments. H. Z. and M. L. conducted the experiments and carried out the data analysis. T. Y., R. M., and Y. Z. performed and analyzed the *in vitro* and *in vivo* experiments. F. L. provided essential technical support and performed additional data validation. J. L. and X. W. directed this study and refined the manuscript. All contributors provided critical revisions and endorsed the final version.

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Data availability The data that support the findings of this study are available in the supplementary information of this article.

Declarations

Conflict of interest No competing interests declared.

Ethics approval All the research protocols and animal work in this study and the experiments involving human subjects were approved by the First Affiliated Hospital, Sun Yat-sen University (Reference No. IIT-2023–207) and the Animal Ethics Committee of Sun Yat-sen University (Approval No. SYSU-IACUC-2024–000001). All human participants provided written informed consent prior to enrollment.

Consent for publication Not required.

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